

9. For a particular set of reaction conditions, the actual yield is the quantity of desired product made in a reaction.

Some examples of reactions with their desired products are shown.

Equation	Desired product
$\text{Ba}(\text{NO}_3)_2(\text{aq}) + \text{Na}_2\text{SO}_4(\text{aq}) \rightarrow \text{BaSO}_4(\text{s}) + 2\text{NaNO}_3(\text{aq})$	$\text{BaSO}_4(\text{s})$
$\text{CH}_3\text{OH}(\ell) + \text{C}_2\text{H}_5\text{COOH}(\ell) \rightleftharpoons \text{C}_2\text{H}_5\text{COOCH}_3(\ell) + \text{H}_2\text{O}(\ell)$	$\text{C}_2\text{H}_5\text{COOCH}_3(\ell)$
$\text{Mg}(\text{s}) + 2\text{HCl}(\text{aq}) \rightarrow \text{MgCl}_2(\text{aq}) + \text{H}_2(\text{g})$	$\text{H}_2(\text{g})$

Using your knowledge of chemistry, describe how the actual yield in a reaction could be determined.

Your answer should include experimental procedures that could be used to determine the quantity of product made in reactions such as the examples shown in the table.

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9. (continued)

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